The tethered bracket is metriplectic, not Poisson: $\{F, F\}_{\text{tethered}} = T(F, F) \neq 0$

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4 November 2026

Abstract

We record, with precise terminology, why the classical Lie–Poisson bracket is a Poisson bracket whose symplectic leaves are the coadjoint orbits, and why the tethered sum $\{F, G\}_{\text{tethered}} = \{F, G\}_{\text{LP}} + T(F, G)$, with T the minimal metric (Frohmanian) tether, is a metriplectic bracket that is not Poisson and has no symplectic foliation in the Poisson sense. We give the verified identity $\{F, F\}_{\text{tethered}} = T(F, F) = -\kappa w \|\Pi_u \delta F\|^2$, an explicit example where it equals $-\kappa w \neq 0$, a SymPy check, and a Lean 4/Mathlib proof using only the standard axioms. This note makes no claim about Navier–Stokes regularity or the Clay Millennium Problem.

1 The classical Lie–Poisson bracket

For a Lie algebra $\mathfrak{g}$ and $F, G \in C^\infty(\mathfrak{g}^*)$, the (minus) Lie–Poisson bracket is $\{F, G\}_{\text{LP}}(\mu) = -\langle \mu, [\delta F/\delta \mu, \delta G/\delta \mu] \rangle$. It is bilinear, antisymmetric, satisfies the Leibniz rule and the Jacobi identity (the latter inherited from the Jacobi identity of $[\cdot, \cdot]$). These properties are exactly what make it a Poisson bracket. For a Poisson bracket the Hamiltonian vector fields span an integrable generalized distribution whose leaves are symplectic; for the Lie–Poisson bracket these symplectic leaves are the coadjoint orbits, with the Kirillov–Kostant–Souriau form. Antisymmetry gives $\{F, F\}_{\text{LP}} = 0$; Casimirs (functions constant on coadjoint orbits) are central. Example: $\mathfrak{g} = \mathfrak{so}(3)$, $\mathfrak{g}^* \cong \mathbb{R}^3$, $\{F, G\}_{\text{LP}}(m) = -m \cdot (\nabla F \times \nabla G)$. The leaves are the spheres $|m| = \text{const}$ (and the origin), and $C = |m|^2$ is a Casimir.

2 The tether and the tethered sum

The source definition (repository `frohmanian-symplectic-tether-may-form`, file `TetherMayForm/Basic.lean`, definitions `frohmanianTether`, `tetheredBracket`) is

$$T(F, G) = -\kappa\, w\, \langle \Pi_u\, \delta F,\ \Pi_u\, \delta G \rangle, \qquad \Pi_u x = x - \frac{\langle u, x \rangle}{\|u\|^2} u,$$

the L^2-orthogonal projection off $\text{span}\{u\}$, with $u = \delta H$ the velocity (gradient of the kinetic energy), w standing for $\int |\omega|^2$ (collapsed to a scalar $w \geq 0$ in the formal model) and $\kappa > 0$. The tethered sum is $\{F, G\}_{\text{tethered}} = \{F, G\}_{\text{LP}} + T(F, G)$. T is symmetric, negative semidefinite for $\kappa, w \geq 0$, and degenerate on energy ($T(F, H) = 0$ since $\Pi_u u = 0$). A sum of an antisymmetric Poisson bracket and a symmetric, semidefinite bracket degenerate on the Hamiltonian is a

metriplectic bracket (in this model the entropy-type compatibility $\{S, \cdot\}_{\mathrm{LP}} = 0$ is not part of the definition).

Proposition 1 (verified)**.** *For every F,*
$\{F, F\}_{\mathrm{tethered}} = \{F, F\}_{\mathrm{LP}} + T(F, F) = T(F, F) = -\kappa w \|\Pi_u \delta F\|^2$*. In particular, on $\mathfrak{so}(3)^*$ with $u = e_1$ and $F(m) = m_2$ ($\delta F = e_2 \perp u$), $\{F, F\}_{\mathrm{tethered}} = -\kappa w$, which is nonzero whenever $\kappa w \neq 0$ (e.g. $\kappa = 3/2$, $w = 2$ gives -3).*

Proof. $\{F, F\}_{\mathrm{LP}} = 0$ by antisymmetry; $\Pi_u e_2 = e_2$, so $T(F, F) = -\kappa w \|e_2\|^2 = -\kappa w$. □

3 Consequences, stated carefully

- Since $\{F, F\}_{\mathrm{tethered}} \neq 0$, the tethered sum is not antisymmetric, hence not a Poisson bracket. Its generator $F \mapsto \{F, H\}_{\mathrm{tethered}}$ need not be a Hamiltonian vector field, and it does not define a symplectic foliation in the Poisson sense; in particular the coadjoint orbits are not its symplectic leaves.
- Casimirs of the Lie–Poisson part are not central for the tethered sum: with $C = |m|^2$, $\{m_2, C\}_{\mathrm{tethered}} = T(m_2, C) = -2\kappa w\, m_2 \neq 0$ in general. So the dissipative part can move states across coadjoint orbits.
- The Jacobi identity also fails for the tethered sum when $\kappa w \neq 0$ (explicit witness `tethered_jacobiator_ne_zero` in `JacobiWitness.lean` of the same repository).

Remark 1. In the formal model, u is a fixed vector and w a scalar; the curl/Biot–Savart relation and the spatial weight $|\omega|^2$ are not modelled. Nothing here concerns regularity of Navier–Stokes.

4 Verification

`verify_self_bracket.py` (SymPy) checks antisymmetry and Jacobi of $\{\cdot, \cdot\}_{\mathrm{LP}}$ on $\mathfrak{so}(3)^*$, centrality of $|m|^2$, $\{m_2, m_2\}_{\mathrm{tethered}} = -\kappa w$, and $\{m_2, |m|^2\}_{\mathrm{tethered}} = -2\kappa w m_2$.
`TetherMayForm/TetheredSelfBracket.lean` proves, for any alternating classical form P, `tetheredBracket_self_eq`: $\{F, F\}_{\mathrm{tethered}} = -\kappa w \|\Pi_u F\|^2$; `witness`: the $\mathbb{R}^3$ value $-\kappa w$; and `tetheredBracket_not_alternating` for $\kappa w \neq 0$. `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`.